

Indexing Strategy

Basis Data (IF202)

Optimasi Query

Indexing

Index adalah struktur data (B-tree, hash) yang mempercepat pencarian, seperti indeks di buku. Trade-off: mempercepat SELECT, memperlambat INSERT/UPDATE.

Jenis Index

Tipe	Struktur	Use Case
B-Tree	Balanced tree	Range query, ORDER BY
Hash	Hash table	Equality (=)
GIN	Inverted index	Full-text, JSONB, array
Composite	Multi-column	Query dengan beberapa kolom

```
CREATE INDEX idx_students_course ON students(course_id);
CREATE INDEX idx_enrollment_composite ON enrollment(course_id, student_id);
CREATE INDEX idx_name_trgm ON users USING gin(name gin_trgm_ops);
```

Ringkasan

Index kolom yang sering di-WHERE, JOIN, ORDER BY. Composite index ikuti leftmost prefix rule. Jangan over-index: setiap index memperlambat write.